JAGADISH JANAKIRAMAN

Performance Engineering Lead | Banking & Healthcare

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EDUCATION

Master of Science, Data Science — University of Arizona, USA
Bachelor of Electronics & Communications Engineering — University of Madras, India

CERTIFICATIONS

PMP · Microsoft Azure Fundamentals · Certified SAFe Agilist · Certified Scrum Professional · Dynatrace Associate · Splunk User · ITIL Foundation · HP Certified LoadRunner Specialist · Sun Certified Java Programmer

PROFESSIONAL SUMMARY

Performance Engineering Lead with 20+ years across banking and healthcare and a track record of owning performance strategy for complex digital programs. Experienced in defining program-level SLAs/KPIs, building governance frameworks, coordinating performance testing across multi-vendor agile workstreams, and providing executive-level reporting on system readiness. Hands-on across Web (Selenium), Middleware/API layers, and Core Banking platforms — with deep expertise in JMeter, LoadRunner, Dynatrace (11 years), and CI/CD pipeline integration. Led performance for the Kaiser Permanente Pharmacy Modernisation (10x scale across 400+ stores) and managed a 30-member team at Accenture; led performance engineering for TD Bank's core banking migration and now its document verification services. PMP- and SAFe-certified.

CORE COMPETENCIES

Program Strategy	Enterprise performance framework design, SLA/KPI/TPS definition, workload modelling, performance risk register, governance across multi-vendor programs
Load Testing	JMeter, LoadRunner Enterprise & Cloud — Web, Mobile, API, and Core Banking workloads
Mobile & Web	Selenium (incl. an in-house JavaScript Selenium framework), Puppeteer, Chrome mobile emulation — web and mobile browser performance testing, multi-channel end-to-end transaction flow coverage
API & Middleware	REST/HTTP, SOAP, Postman, MuleSoft API layer testing, service virtualisation — CA DevTest, Ready API
APM & Observability	Dynatrace, Splunk, Datadog, Grafana, InfluxDB
Cloud & Container	Azure (AKS, App Services, Monitor), Docker, Kubernetes — HPA tuning, pod scaling, resource limits
CI/CD & DevOps	GitHub Actions, Jenkins — performance test automation with threshold-based pass/fail enforcement; GitHub Actions workflows built for on-demand execution
Test Strategy	NFR definition, p95/p99 profiling, Micro-benchmark, Breakpoint, Endurance, Soak, Spike, Chaos/Failover (Chaos Monkey), Operability, Negative, Monitoring, Fallback, Fall-forward
Databases	Oracle, PostgreSQL, MongoDB, Cosmos DB — query plan analysis & tuning
Languages	Java (Spring Boot stub services), JavaScript / Node.js, Python, Shell Scripting, SQL
AI-Assisted Engineering	GitHub Copilot (test scripts, test data, and y4m camera-feed video generation), Ollama (local LLM analysis of test results)
Reporting	Power BI, Tableau — executive performance scorecards, dashboards, bottleneck heatmaps and trend analysis
Leadership	PMP, SAFe Agilist, Scrum — executive reporting, cross-functional teams, vendor management, mentoring

PROFESSIONAL EXPERIENCE

Performance Engineering Lead · TD Bank (Toronto-Dominion Bank)

Toronto, ON | November 2022 – Present

Strategy & Governance

- Acted as the single point of accountability for performance engineering on TD Bank's core banking migration program (2022–2024) — a multi-workstream transformation serving millions of retail and commercial banking customers. Since 2024, leading performance testing for the bank's document verification services.
- Defined program-wide SLAs, KPIs, TPS targets, and latency thresholds for customer-facing Web and API endpoints; established governance processes covering workload modelling, test data management, environment readiness, and defect triage across agile workstreams.
- Built and maintained a performance risk register tracking program-level risks, bottleneck trends, and mitigation actions — reviewed with enterprise architecture and infrastructure stakeholders at each release milestone.

Execution & Innovation

- Pioneered performance testing for TD Bank's document verification service — a first for the bank — by engineering a framework that injects live camera feed inputs (document capture and selfie capture) using y4m video files at runtime, enabling realistic load simulation of a media-intensive Web/Mobile workflow that no existing tooling could handle; later cut test-video generation from hours to minutes with a GitHub Copilot agent.
- Built UI performance scripts on the bank's in-house JavaScript/Selenium framework, using GitHub Copilot to generate scripts from recorded Puppeteer flows instead of hand-writing XPaths.
- Built GitHub Actions workflows to run JMeter performance test suites, enabling on-demand execution with automated threshold enforcement for response time and error rate validation — establishing the foundation for full CI/CD pipeline integration.
- Ran chaos testing scenarios using TD's in-house framework — dependency failures, network latency injection, pod termination on AKS — validating resilience of core banking microservices before each production release.
- Identified a critical HPA misconfiguration through spike testing after production showed surges to ~2,000 TPS at the top of every hour against a 700 TPS tested peak; lowered the scaling threshold with the infrastructure team and validated corrected autoscaling behaviour.
- Right-sized ~40 Azure App Service apps on the core banking platform after utilization and capacity testing, saving about CAD 50,000 per month.

Observability & Stakeholder Reporting

- Defined SLI/SLO targets for customer-facing microservices; built Dynatrace dashboards tracking p95/p99 latency and throughput — providing release teams with data-driven go/no-go signals at every deployment milestone.
- Provided executive-level performance scorecards and bottleneck heatmaps at program milestone reviews, translating system behaviour data into business-aligned readiness recommendations.
- Built Splunk queries and dashboards correlating application logs and infrastructure metrics to support faster triage during high-severity incidents across the core banking platform.

Performance Engineering Lead / Manager · Kaiser Permanente (engaged via Accenture, then DXC Technology)

San Francisco, CA | August 2011 – October 2022

Program Orchestration & Governance

- Led performance engineering as the single point of accountability across Kaiser's Pharmacy Modernisation program — coordinating test planning and execution across multiple vendor teams, agile workstreams, and enterprise architecture stakeholders.
- Scaled the Kaiser Pharmacy platform 10x across 400+ stores within 24 months; tuned systems to sustain 3x peak demand during COVID-19 by proactively running Breakpoint and Soak tests before demand surges hit production.
- Defined multi-layered test strategies covering Web, API middleware, and backend database layers — including REST/SOAP API performance testing across distributed microservices with parameterised data and SLO-aligned p95 latency targets.
- Aligned with Enterprise Architecture and Infrastructure teams on scalability planning and capacity needs; validated cloud migration performance from on-premises to Azure and IBM Bluemix using production transaction traces as load models.

Test Automation & CI/CD

- Designed a Video Conference Performance Testing Framework using LoadRunner, Selenium (Web), Java, and Power BI dashboards — providing executive-ready coverage across end-to-end latency, error rate, and throughput for 400+ stores.
- Integrated JMeter and LoadRunner into Jenkins CI/CD pipelines, embedding non-functional regression testing into sprint releases and catching performance defects earlier in the delivery cycle.

- Introduced dynamic test data provisioning via LoadRunner Virtual Table Server (VTS), eliminating data-starvation bottlenecks and enabling 24/7 uninterrupted endurance runs; developed reusable Citrix ICA scripting strategies reducing effort by ~60%.

Observability & Mentoring

- Used Dynatrace distributed tracing across pharmacy microservices to triage performance issues quickly through method-level hotspot analysis and transaction flow visualisation.
- Established metrics-based release gates using pre/post-test statistical analysis — adopted as the standard release qualification process across the program and presented at executive milestone reviews.
- Mentored and guided Performance Test Leads and Engineers across streams; defined best practices, reusable accelerators, and scripting standards adopted across multiple vendor teams; at Accenture, managed a 30-member performance engineering team in India alongside onshore US teams.

Performance Test Engineer · Tata Consultancy Services (TCS)

India & USA | April 2004 – August 2011

- Delivered performance testing engagements for Deluxe Corporation, Citi Bank, and Bank of America — resolved throughput and latency bottlenecks on high-volume transactional banking applications to meet business SLAs; built reusable scripting strategies adopted across multiple client accounts.
- Mentored engineers in scripting, scenario design, and results analysis across concurrent US and India engagements, building team capability that scaled delivery without adding headcount.

PROJECTS

- Performance Testing Framework (github.com/jagsrj/perf-framework) — self-hosted JMeter console built with Node.js, Docker, PostgreSQL, InfluxDB and Grafana; versioned scripts, automatic baseline comparison, and local AI result analysis with Ollama.